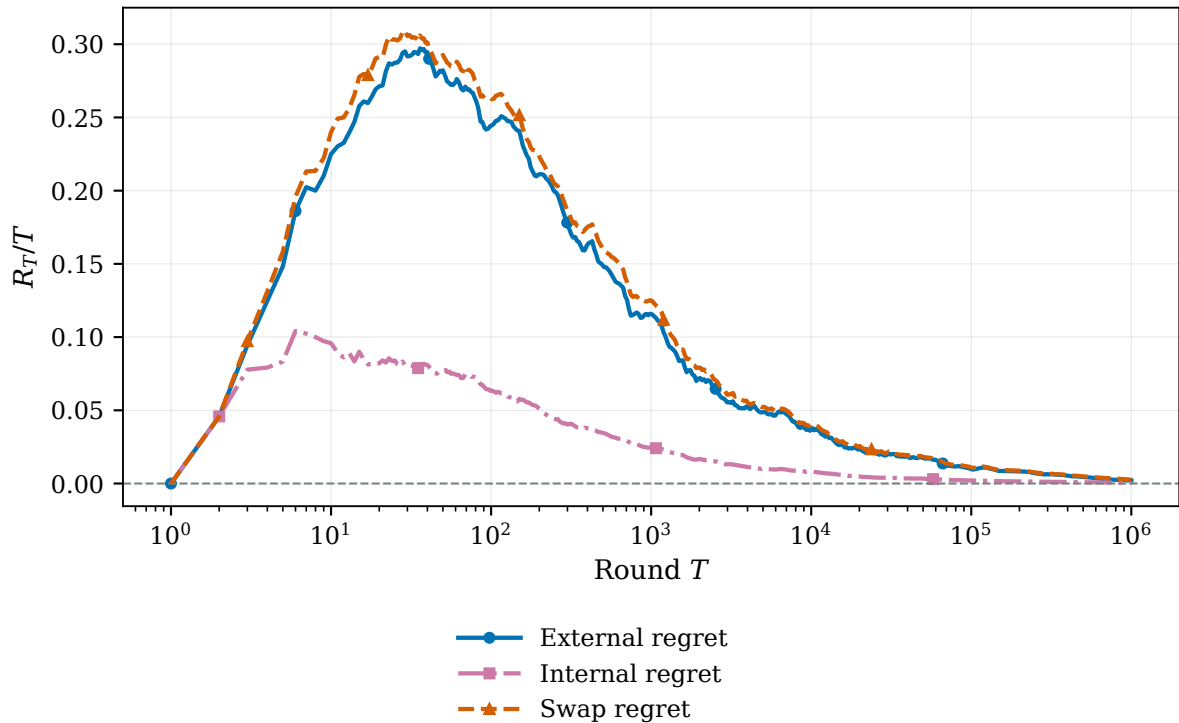


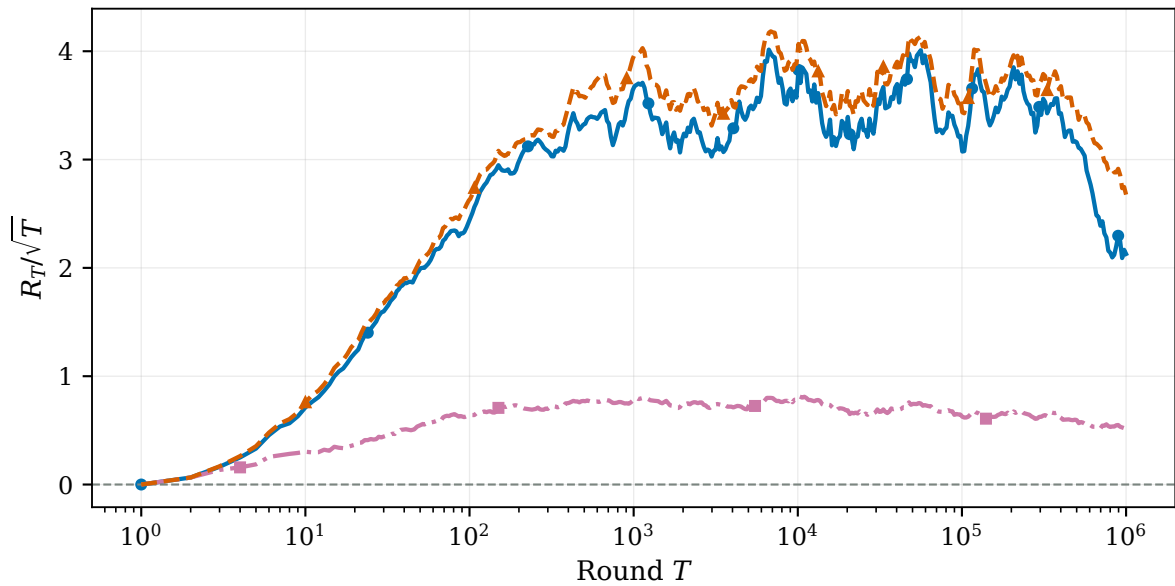
Experiment Information

Figure:	Regret-notion comparison
Environment:	Lazy random walk
Actions:	10
Feedback:	Full information
Algorithm:	BM-Hedge
Horizon:	1,000,000
Replicates:	12
Seed:	42
Regrets:	External, Internal, Swap
View:	R / T
Final endpoints at T:	1,000,000
Endpoint statistics:	replicate mean, sample SD (ddof=1), SE=SD/sqrt(n)
External:	mean = 0.00212853, SD = 0.000752986, SE = 0.000217368
Internal:	mean = 0.000495717, SD = 8.48667e-05, SE = 2.44989e-05
Swap:	mean = 0.00267633, SD = 0.000439674, SE = 0.000126923



Experiment Information

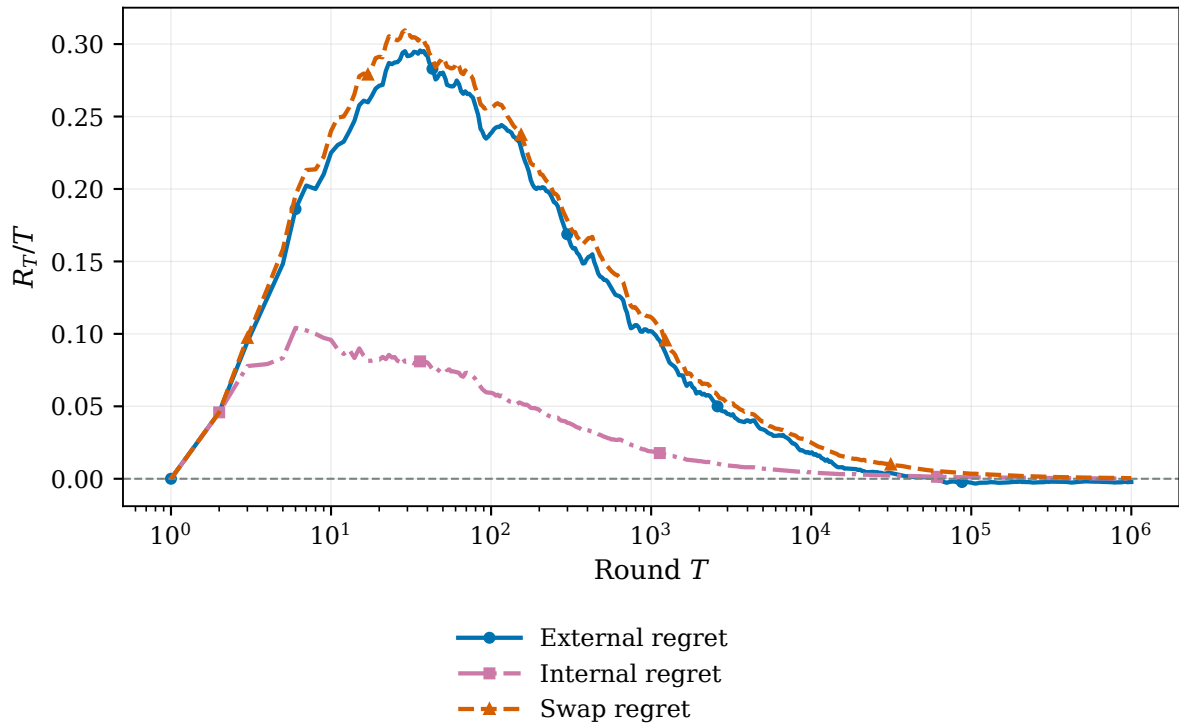
Figure:	Regret-notion comparison
Environment:	Lazy random walk
Actions:	10
Feedback:	Full information
Algorithm:	BM-Hedge
Horizon:	1,000,000
Replicates:	12
Seed:	42
Regrets:	External, Internal, Swap
View:	$R / \sqrt{T}$
Final endpoints at T:	1,000,000
Endpoint statistics:	replicate mean, sample SD (ddof=1), SE=SD/sqrt(n)
External:	mean = 2.12853, SD = 0.752986, SE = 0.217368
Internal:	mean = 0.495717, SD = 0.0848667, SE = 0.0244989
Swap:	mean = 2.67633, SD = 0.439674, SE = 0.126923



- External regret
- Internal regret
- Swap regret

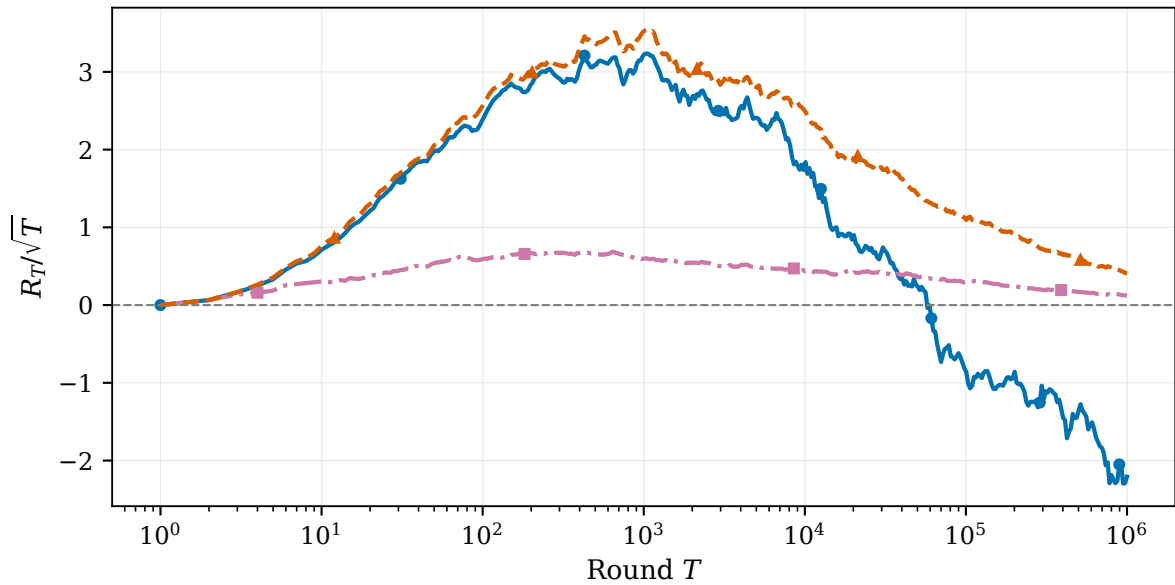
Experiment Information

Figure:	Regret-notion comparison
Environment:	Lazy random walk
Actions:	10
Feedback:	Full information
Algorithm:	BM-OptHedge
Horizon:	1,000,000
Replicates:	12
Seed:	42
Regrets:	External, Internal, Swap
View:	R / T
Final endpoints at T:	1,000,000
Endpoint statistics:	replicate mean, sample SD (ddof=1), SE=SD/sqrt(n)
External:	mean = -0.00220859, SD = 0.00143739, SE = 0.00041494
Internal:	mean = 0.000121275, SD = 4.25657e-05, SE = 1.22877e-05
Swap:	mean = 0.000410117, SD = 6.23952e-05, SE = 1.80119e-05



Experiment Information

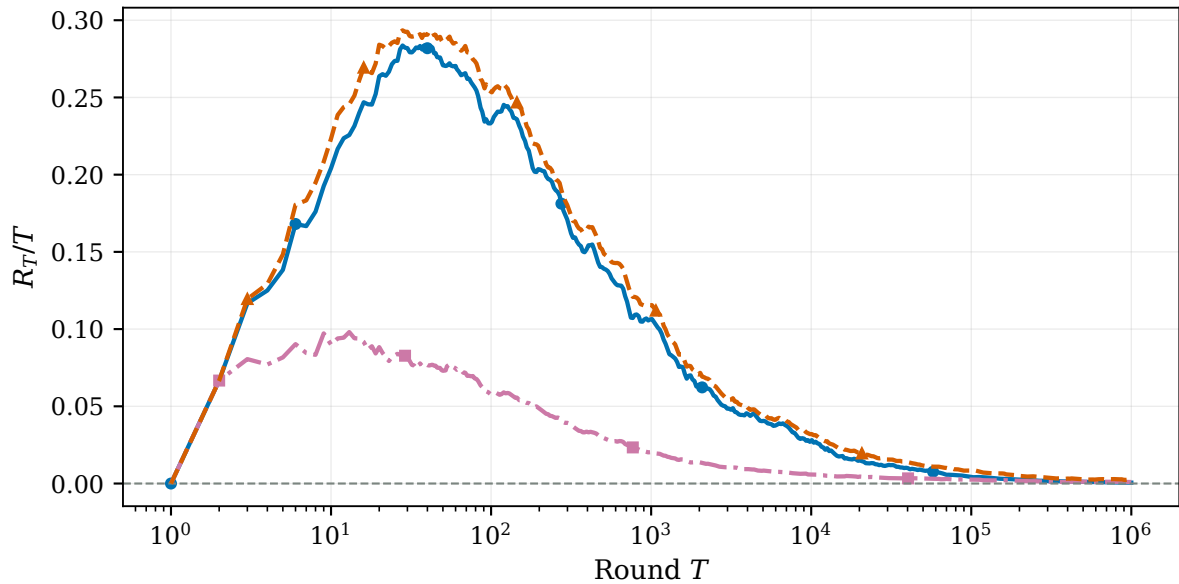
Figure:	Regret-notion comparison
Environment:	Lazy random walk
Actions:	10
Feedback:	Full information
Algorithm:	BM-OptHedge
Horizon:	1,000,000
Replicates:	12
Seed:	42
Regrets:	External, Internal, Swap
View:	$R / \sqrt{T}$
Final endpoints at T:	1,000,000
Endpoint statistics:	replicate mean, sample SD (ddof=1), SE=SD/sqrt(n)
External:	mean = -2.20859, SD = 1.43739, SE = 0.41494
Internal:	mean = 0.121275, SD = 0.0425657, SE = 0.0122877
Swap:	mean = 0.410117, SD = 0.0623952, SE = 0.0180119



- External regret
- Internal regret
- Swap regret

Experiment Information

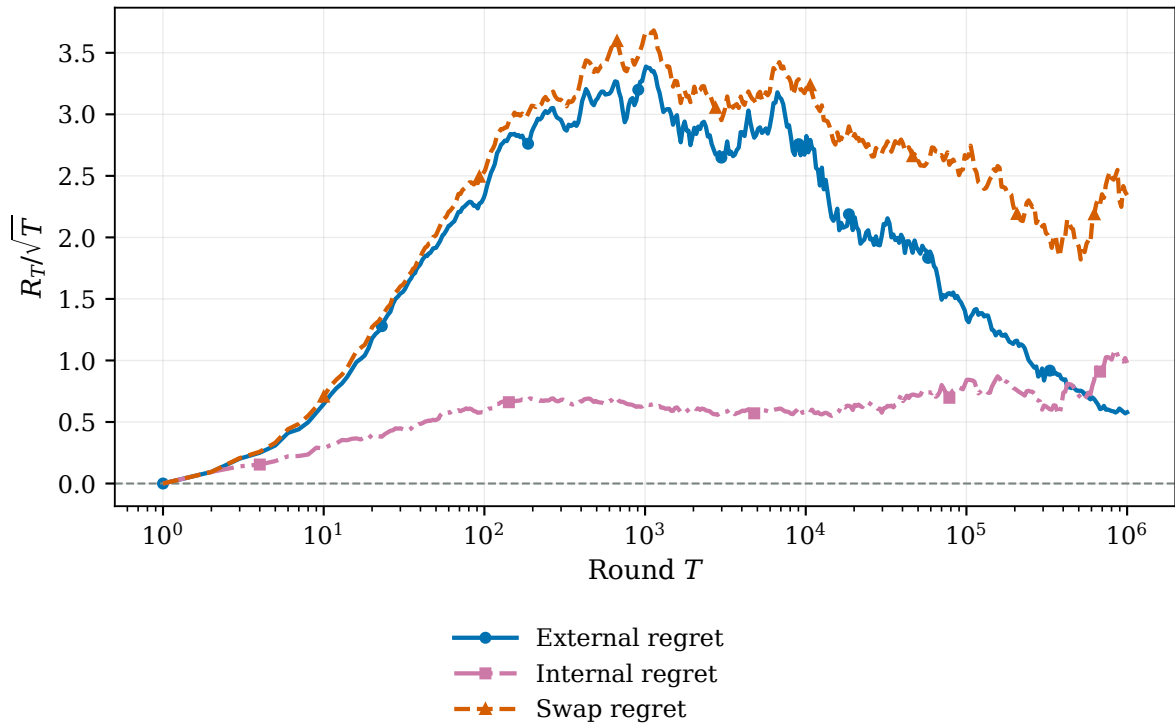
Figure:	Regret-notion comparison
Environment:	Lazy random walk
Actions:	10
Feedback:	Full information
Algorithm:	Hedge
Horizon:	1,000,000
Replicates:	12
Seed:	42
Regrets:	External, Internal, Swap
View:	R / T
Final endpoints at T:	1,000,000
Endpoint statistics:	replicate mean, sample SD (ddof=1), SE=SD/sqrt(n)
External:	mean = 0.000578758, SD = 0.0001043, SE = 3.01089e-05
Internal:	mean = 0.000979592, SD = 0.000657893, SE = 0.000189917
Swap:	mean = 0.00234382, SD = 0.00112066, SE = 0.000323507



- External regret
- Internal regret
- Swap regret

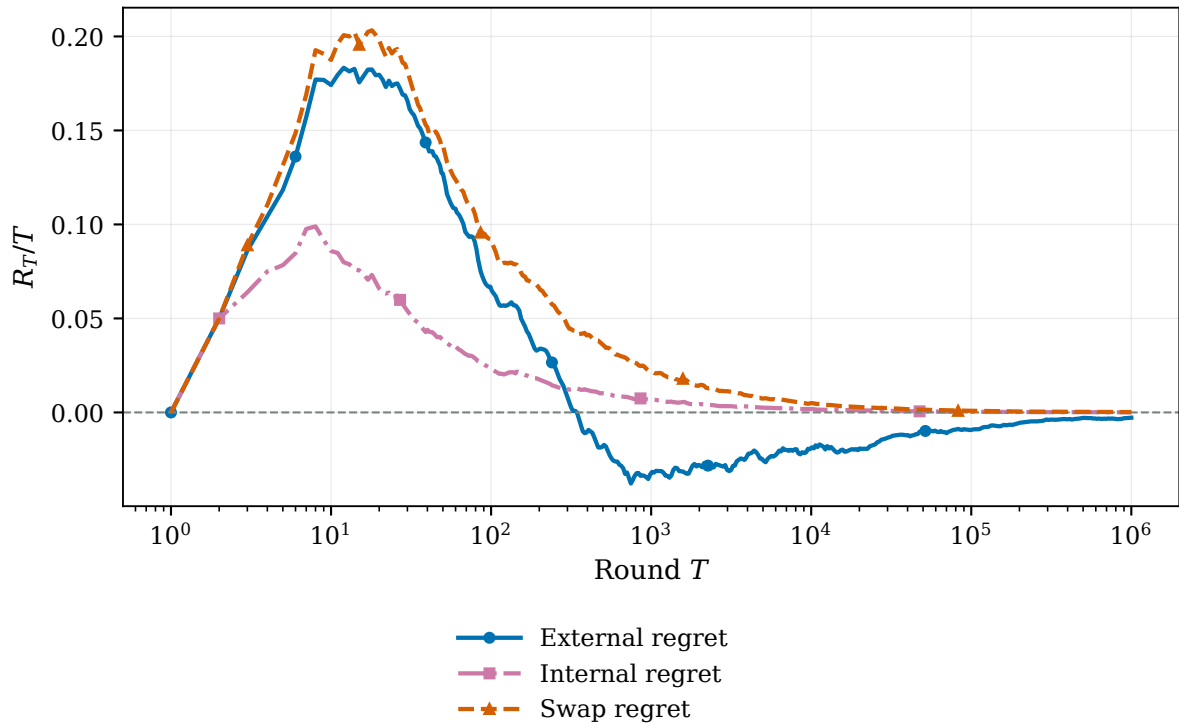
Experiment Information

Figure:	Regret-notion comparison
Environment:	Lazy random walk
Actions:	10
Feedback:	Full information
Algorithm:	Hedge
Horizon:	1,000,000
Replicates:	12
Seed:	42
Regrets:	External, Internal, Swap
View:	$R / \sqrt{T}$
Final endpoints at T:	1,000,000
Endpoint statistics:	replicate mean, sample SD (ddof=1), SE=SD/sqrt(n)
External:	mean = 0.578758, SD = 0.1043, SE = 0.0301089
Internal:	mean = 0.979592, SD = 0.657893, SE = 0.189917
Swap:	mean = 2.34382, SD = 1.12066, SE = 0.323507



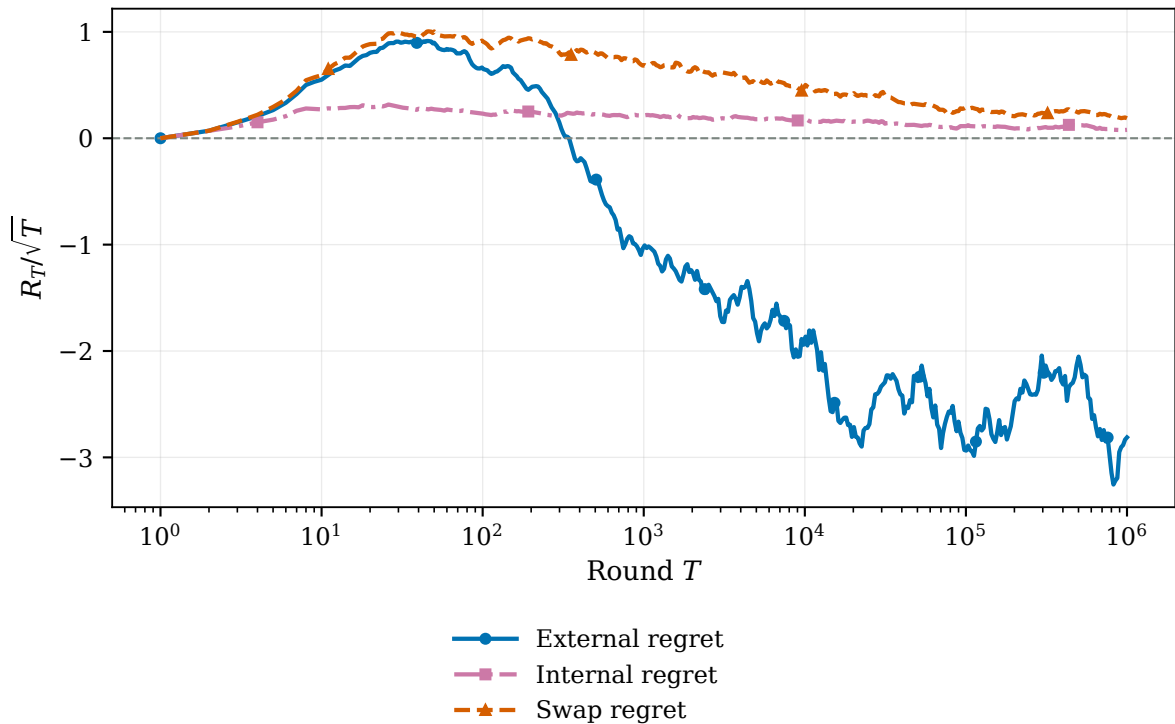
Experiment Information

Figure:	Regret-notion comparison
Environment:	Lazy random walk
Actions:	10
Feedback:	Full information
Algorithm:	Ito-Hedge
Horizon:	1,000,000
Replicates:	12
Seed:	42
Regrets:	External, Internal, Swap
View:	R / T
Final endpoints at T:	1,000,000
Endpoint statistics:	replicate mean, sample SD (ddof=1), SE=SD/sqrt(n)
External:	mean = -0.00281221, SD = 0.00150603, SE = 0.000434753
Internal:	mean = 7.895e-05, SD = 4.12671e-05, SE = 1.19128e-05
Swap:	mean = 0.000187867, SD = 9.92245e-05, SE = 2.86437e-05



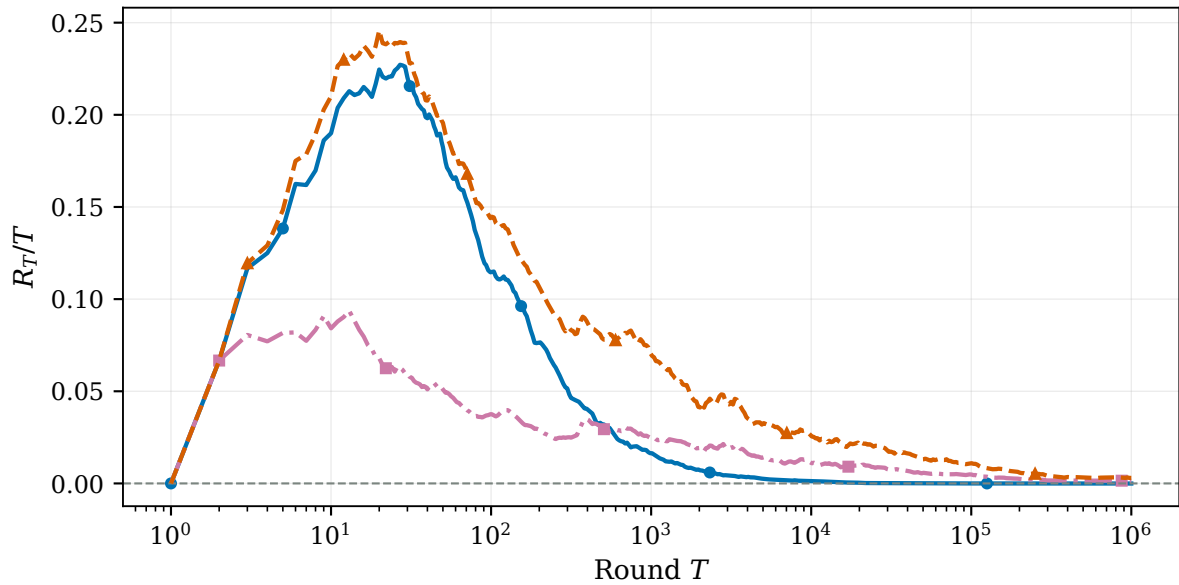
Experiment Information

Figure: Regret-notion comparison
Environment: Lazy random walk
Actions: 10
Feedback: Full information
Algorithm: Ito-Hedge
Horizon: 1,000,000
Replicates: 12
Seed: 42
Regrets: External, Internal, Swap
View: $R / \sqrt{T}$
Final endpoints at T: 1,000,000
Endpoint statistics: replicate mean, sample SD (ddof=1), SE=SD/sqrt(n)
External: mean = -2.81221, SD = 1.50603, SE = 0.434753
Internal: mean = 0.07895, SD = 0.0412671, SE = 0.0119128
Swap: mean = 0.187867, SD = 0.0992245, SE = 0.0286437



Experiment Information

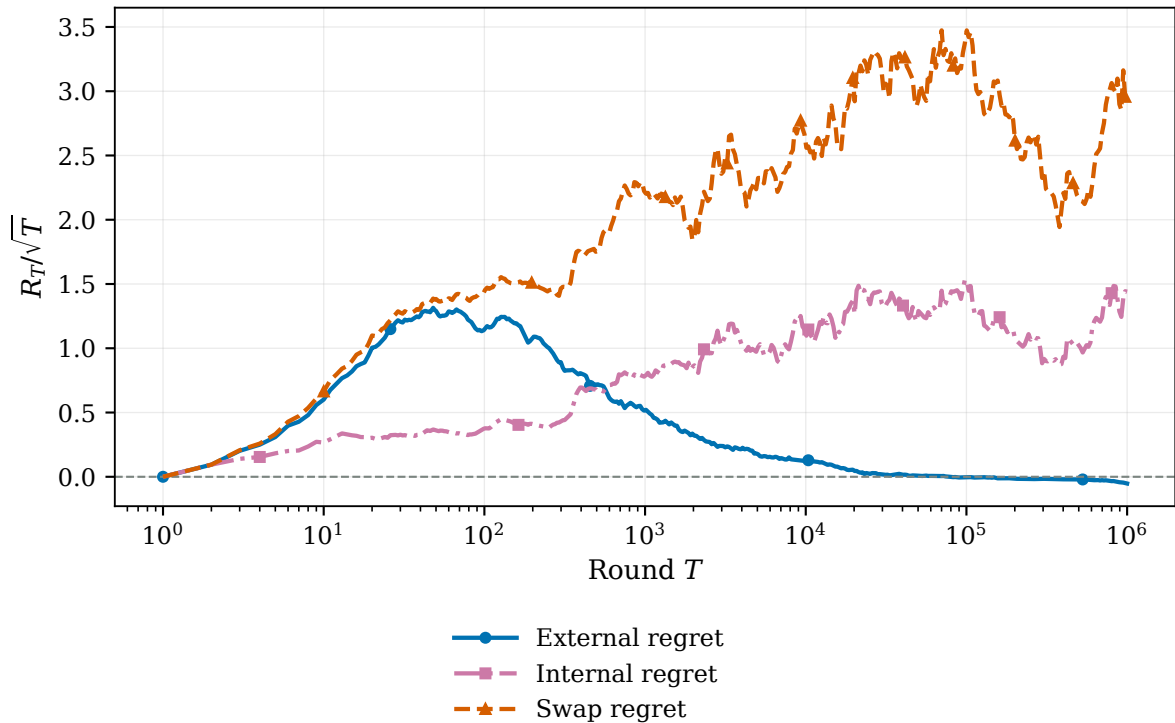
Figure:	Regret-notion comparison
Environment:	Lazy random walk
Actions:	10
Feedback:	Full information
Algorithm:	OptHedge
Horizon:	1,000,000
Replicates:	12
Seed:	42
Regrets:	External, Internal, Swap
View:	R / T
Final endpoints at T:	1,000,000
Endpoint statistics:	replicate mean, sample SD (ddof=1), SE=SD/sqrt(n)
External:	mean = -5.24417e-05, SD = 4.36894e-05, SE = 1.2612e-05
Internal:	mean = 0.00144024, SD = 0.000797076, SE = 0.000230096
Swap:	mean = 0.00298063, SD = 0.00159729, SE = 0.000461098



- External regret
- Internal regret
- Swap regret

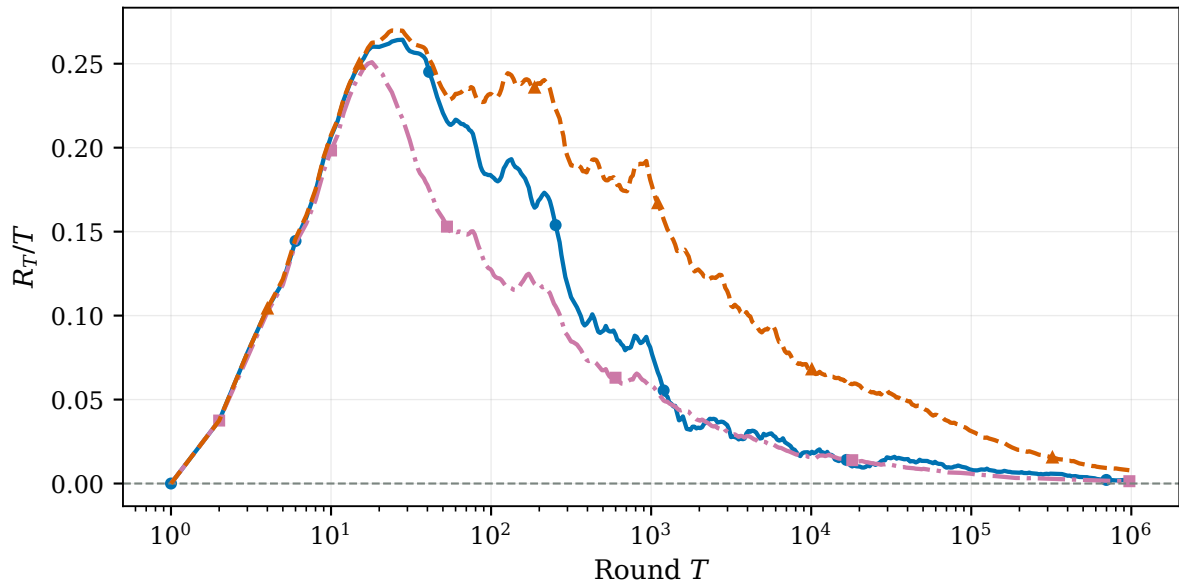
Experiment Information

Figure:	Regret-notion comparison
Environment:	Lazy random walk
Actions:	10
Feedback:	Full information
Algorithm:	OptHedge
Horizon:	1,000,000
Replicates:	12
Seed:	42
Regrets:	External, Internal, Swap
View:	$R / \sqrt{T}$
Final endpoints at T:	1,000,000
Endpoint statistics:	replicate mean, sample SD (ddof=1), SE=SD/sqrt(n)
External:	mean = -0.0524417, SD = 0.0436894, SE = 0.012612
Internal:	mean = 1.44024, SD = 0.797076, SE = 0.230096
Swap:	mean = 2.98063, SD = 1.59729, SE = 0.461098



Experiment Information

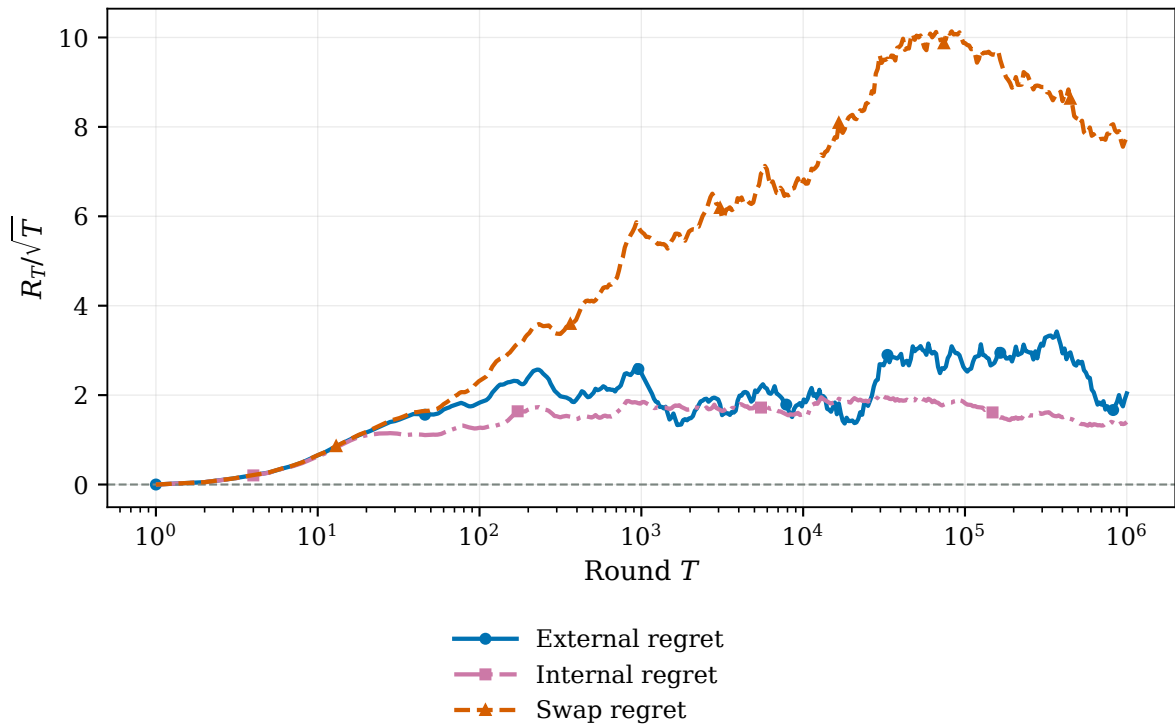
Figure:	Regret-notion comparison
Environment:	Lazy random walk
Actions:	10
Feedback:	Full information
Algorithm:	RM
Horizon:	1,000,000
Replicates:	12
Seed:	42
Regrets:	External, Internal, Swap
View:	R / T
Final endpoints at T:	1,000,000
Endpoint statistics:	replicate mean, sample SD (ddof=1), SE=SD/sqrt(n)
External:	mean = 0.00203804, SD = 0.00230171, SE = 0.000664446
Internal:	mean = 0.0013993, SD = 0.000615364, SE = 0.00017764
Swap:	mean = 0.00776705, SD = 0.00323116, SE = 0.000932755



- External regret
- Internal regret
- Swap regret

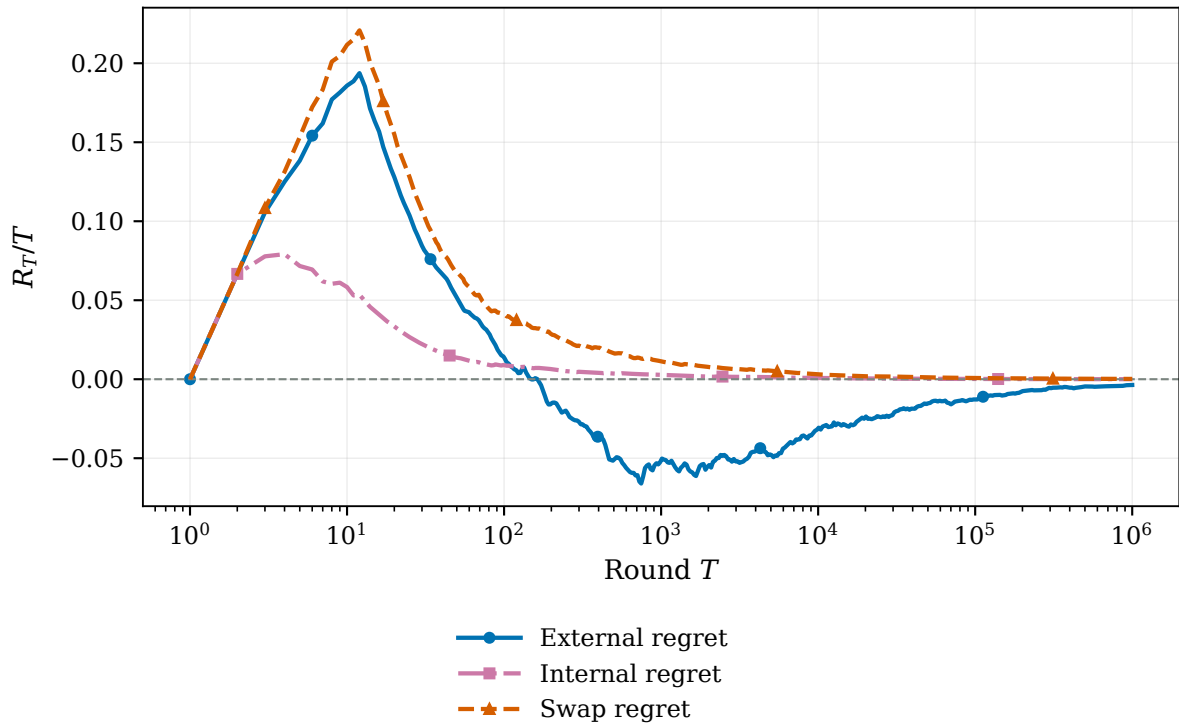
Experiment Information

Figure:	Regret-notion comparison
Environment:	Lazy random walk
Actions:	10
Feedback:	Full information
Algorithm:	RM
Horizon:	1,000,000
Replicates:	12
Seed:	42
Regrets:	External, Internal, Swap
View:	$R / \sqrt{T}$
Final endpoints at T:	1,000,000
Endpoint statistics:	replicate mean, sample SD (ddof=1), SE=SD/sqrt(n)
External:	mean = 2.03804, SD = 2.30171, SE = 0.664446
Internal:	mean = 1.3993, SD = 0.615364, SE = 0.17764
Swap:	mean = 7.76705, SD = 3.23116, SE = 0.932755



Experiment Information

Figure:	Regret-notion comparison
Environment:	Lazy random walk
Actions:	10
Feedback:	Full information
Algorithm:	SRM
Horizon:	1,000,000
Replicates:	12
Seed:	42
Regrets:	External, Internal, Swap
View:	R / T
Final endpoints at T:	1,000,000
Endpoint statistics:	replicate mean, sample SD (ddof=1), SE=SD/sqrt(n)
External:	mean = -0.00364579, SD = 0.00159322, SE = 0.000459922
Internal:	mean = 4.0175e-05, SD = 1.14256e-05, SE = 3.29828e-06
Swap:	mean = 0.000185383, SD = 7.08558e-05, SE = 2.04543e-05



Experiment Information

Figure: Regret-notion comparison
Environment: Lazy random walk
Actions: 10
Feedback: Full information
Algorithm: SRM
Horizon: 1,000,000
Replicates: 12
Seed: 42
Regrets: External, Internal, Swap
View: $R / \sqrt{T}$
Final endpoints at T: 1,000,000
Endpoint statistics: replicate mean, sample SD (ddof=1), SE=SD/sqrt(n)
External: mean = -3.64579, SD = 1.59322, SE = 0.459922
Internal: mean = 0.040175, SD = 0.0114256, SE = 0.00329828
Swap: mean = 0.185383, SD = 0.0708558, SE = 0.0204543

